



BHABHA UNIVERSITY, BHOPAL (M.P.)

Scheme of Examination Programme – Bachelor of Technology Session 2021-22

Semester – VII

Branch – Civil Engineering

S. No.	Subject Code	Subject Name	Periods per week Hours / week.			Total Credits	Maximum Marks (Theory Slot)			Marks for End Sem. Examination		Total Marks
			L	T	P		End Sem. Exam.	Mid Sem Exam	Assignments /Quiz	End Sem. Practical /Viva	Practical Record/ Assignment /Sessional	
1	CE-71	Geotechnical Engg	2	1	2	4	70	20	10	30	20	150
2	CE-72	Structural Design and Drawing (RCC-II)	3	1	-	4	70	20	10	-	-	100
3	CE-73	Environmental Engg-II	3	-	-	3	70	20	10	-	-	100
4	CE-74	Elective-I	2	1	-	3	70	20	10			100
5	CE-75	Modern Construction Technique & Equipment.	2	1	2	4	70	20	10	30	20	150
6	CE-76	Minor Project	-	-	6	3	-	-	-	60	40	100
7	CE-77	Industrial Training (Two weeks)	-	-	6	3	-	-	-	30	20	50
		Total	12	4	16	24	350	100	50	150	100	750

L: Lecture T: Tutorial P: Practical

One credit refers to one hour teaching in Theory, Tutorial and two hour in Practical.

Elective-I	
CE-74 (A) Fluid Mechanics II	CE-74 (C) Advance Pavement Design
CE-74 (B) Cost effective and Eco-friendly structures	CE-74 (D) Intellectual Property rights (IPR)

UNIT –I

INTRODUCTION, INDEX PROPERTIES AND CLASSIFICATION OF SOILS:

Definition and scope of soil mechanics, Origin of soil, formation of soil, clay minerals, Soil structure. 3-phase soil system, Basic terminology, and their relations, index properties of soil - Water content, Field density, Specific gravity, Grain size distribution by sieve and hydrometer analysis, Relative density, Atterberg limits and their determination, Various indices -Flow-Index, Plasticity Index, Toughness Index, Liquidity Index, Activity Ratio.

Different Systems of Soil Classification - Particle Size, Textural, Unified, HRB and IS classification. Field identification of soils.

UNIT –II

PERMEABILITY, EFFECTIVE STRESS AND SEEPAGE THROUGH SOILS: Types of soil water, Capillarity in soils, Flow of water through soils, Darcy's Law, permeability, Factors affecting permeability, Laboratory & field tests for determination of coefficient of permeability, Permeability of layered soils. Seepage pressure, total, Neutral and effective stress, Upward and downward seepage through soils, , Flow nets: characteristics, Methods of construction of flow net, Application of flow net, Quick condition, Laplace Equation for two –dimensional flow, Seepage through anisotropic soil and non-homogenous soil, Seepage through earth dam.

UNIT-III

STRESS DISTRIBUTION IN SOILS AND COMPACTION: Boussinesq's and Westergard's theories for point load, Uniformly loaded circular and rectangular areas, Pressure bulb, Variation of vertical stress under point load along vertical and horizontal plane, Newmark's influence chart for irregular areas. Contact pressure distribution in sands and clays.

Mechanism importance of compaction, Standard Proctor compaction test, Modified compaction test Factors affecting compaction, Effects of compaction on soil properties, Field compaction equipment and compaction quality control.

UNIT-IV

CONSOLIDATION :Types of compressibility, Spring analogy, Immediate settlement, Primary consolidation and secondary consolidation, Stress history of clay, e-p and e-log p curves, Normally consolidated soil, Over consolidated soil and under consolidated soil, Pre-consolidation pressure and its determination, Consolidation test, Terzaghi's 1-D consolidation theory, Coefficient of consolidation, Square root time and Logarithm of time fitting methods, Computation of total settlement.

UNIT-V

SHEAR STRENGTH & STABILIZATION OF SOILS:

SHEAR STRENGTH: Definition and importance of shear strength, Mohr and coulomb failure theories, Mohr's Stress Circles, Measurement of shear strength-Different types of Shear Test namely, Direct Shear Test, Unconfined Compression Test, Tri Axial Compression Test & Vane Shear Test for strength parameters, Strength tests based on drainage conditions, Measurement of pore pressure, Pore pressure parameters,

Strength envelopes, shear strength of sands, Critical void ratio, Liquefaction, Shear strength of clays. Factors affecting shear strength of granular soils and cohesive soils.

STABILIZATION OF SOIL: Introduction, Mechanical stabilization, Cement stabilization, Lime stabilization, Bituminous stabilization, Chemical stabilization, Thermal stabilization, Electrical stabilization, Stabilization by grouting, Use of geo-synthetic materials, Types, Functions and applications of geo-synthetics, Reinforced earth structures-components and construction.

LIST OF EXPERIMENTS:

1. Determination of water content by Oven drying method.
2. Determination of water content by Pycnometer
3. Determination of soil field density by core cutter method
4. Determination of soil field density by sand replacement method
5. Determination of Specific Gravity By Pycnometer.
6. Determination of Consistency Limits (i) Liquid Limit (ii) Plastic Limit (iii) Shrinkage Limit
7. Determination of liquid limit of soil by cone penetrometer.
8. Grain size analysis by sieve shaking method
9. Grain size analysis of fine grained soil by sedimentation using (i) pipette (ii) hydrometer.
10. Determination of coefficient of permeability of soil by- (a) constant head method (b) variable head method.
11. Determination of compaction parameters by- (a) light compaction, (b) heavy compaction.
12. Direct Shear test
13. Triaxial Test
14. Unconfined Compression Strength Test.

Books and References:

1. Punamia B.C., Soil Mechanics & Foundations., Firewall Media, 2017 (16th edition)
2. Alam Singh, Modern Geotechnical Engineering., CBS Publishers & Distributors, 2012 (3rd edition)
3. Gopal Ranjan & ASR Rao, Basic & Applied Soil Mechanics. New Age International, 2016 (3rd edition)
4. S.K Grag, Geotechnical Engineering., Khanna Publishers, 2016 (10th edition)

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject – Structural Design and Drawing (RCC-II)

Sub Code – CE 72

Unit – I

Design of Multistory Buildings - Sway and nonsway buildings, Shear walls and other bracing elements.

Unit II

Design of Earth Retaining Structures: Cantilever and counter fort types retaining walls.

Unit - III

Water Tanks: Tanks on ground and underground tanks: Square, rectangular, circular tanks, Overhead tanks: square, rectangular, circular & intze tanks.

Unit – IV

Design of Silos and Bunkers

Unit - V

T-beam & Slab bridges- for highway loading (IRC Loads). Prestressing concepts materials, systems of prestressing & losses Introduction to working & limit State Design.

Suggested Books: -

1. R.C.C. by O.P. Jain Vol. II
2. R.C.C. by B.C. Punmia
3. Essentials of Bridge engineering – D.J. Victor
4. Bridge Engineering – Ponnuswamy
5. Advanced R.C.C. Design by N.K. RAJU
6. N.Krishna Raju, Prestressed Concrete, Tata Mc Graw Hill, New Delhi.
7. Pre stresses concrete – T.Y. Lin

Unit –I: Unit operations for waste-water treatment

Theory and design of preliminary treatment such as screens, grit chamber, sedimentation and chemical clarification, role of micro-organism in biological treatment.

Unit - II: Biological Treatment of waste-water

Methods of Biological Treatment (Theory & Design) –Trickling Filter, Activated Sludge process (ASP), Oxidation ditch, Septic tank & imhoff tank, theory of sludge.

Unit – III: Advanced Waste-water treatment

Diatomaceous earth filters, Ultrafiltration, Adsorption by activated carbon, Phosphorus removal, Nitrogen removal.

UNIT IV: Introduction of Air pollution

Definition, Sources, classification and characterization of air pollutants. Effects of air pollution on health, vegetation & materials, photochemical smog.

UNIT V: Air pollution chemistry

Meteorological aspects of air pollution dispersion; temperature lapse rate and stability, wind velocity and turbulence, plume behaviour, dispersion of air pollutants, the Gaussian Plume Model.

Reference Books:

1. Water Supply & Sanitary Engg. - G.S. Birdie - Dhanpat Rai Publishing Company, 2. (P) Ltd. New Delhi
3. Waste Water Engg. by B.C. Punmia - Laxmi Publication (P) Ltd. New Delhi
4. Environmental Engg. - M.L. Davis & D.A. Cornwell - Mc Graw Hill Company
5. Chemistry for Environmental Engg. - Sawyer & Mc Carty - Mc Graw Hill Book Company New Delhi
6. Water & Waste Water Technology - Mark J Hammer - Prentice - Hall of India, New Delhi
7. Waste Water Engineering - Metcalf & Eddy - Mc Graw Hill Book Company New Delhi

Elective-I

UNIT-I

Turbulent flow : Laminar and turbulent boundary layers and laminar sublayer, hydrodynamically smooth and rough boundaries, velocity distribution in turbulent flow, resistance of smooth and artificially roughened pipes, commercial pipes, aging of pipes. Pipe flow problems : Losses due to sudden expansion and contraction, losses in pipe fittings and valves, concepts of equivalent length, hydraulic and energy gradient lines, siphon, pipes in series, pipes in parallel, branching of pipes. Pipe Network : *Water Hammer (only quick closure case). transmission of power. *Hardy Cross Method

UNIT-II

Uniform flow in open channels : Channel geometry and elements of channel section, velocity distribution, energy in open channel flow, specific energy, types of flow, critical flow and its computations, uniform flow and its computations, Chezy's and Manning's formulae, determination of normal depth and velocity, Normal and critical slopes, Economical sections, Saint Venet equation.

UNIT-III

Non uniform flow in open channels : Basic assumptions and dynamic equations of gradually varied flow, characteristics analysis and computations of flow profiles, rapidly varied flow hydraulic jump in rectangular channels and its basic characteristics, surges in open channels & channel flow routing, venturi flume.

UNIT-IV

Forces on immersed bodies: Types of drag, drag on a sphere, a flat plate, a cylinder and an aerofoil development of lift, lifting vanes, magnus effect.

UNIT-V

Fluid Machines: Turbines : Classifications, definitions, similarity laws, specific speed and unit quantities, Pelton turbine-their construction and settings, speed regulation, dimensions of various elements, Action of jet, torque, power and efficiency for ideal case, characteristic curves.

Reaction turbines: construction & settings, draft tube theory, runaway speed, simple theory of design and characteristic curves, cavitation. Pumps: Centrifugal pumps : Various types and their important components, manometric head, total head, net positive suction head, specific speed, shut off head, energy losses, cavitation, principle of working and characteristic curves.

Reciprocating pumps: Principle of working, Coefficient of discharge, slip, single acting and double acting pump, Manometric head, Acceleration head.

Reference books:-

1. Fluid Mechanics - Modi & Seth - Standard Book house, Delhi
2. Open Channel Flow by Rangaraju - Tata Mc Graw - Hill Publishing Comp. Ltd., New Delhi
3. Fluid Mechanics - A.K. Jain - Khanna Publishers, Delhi
5. Hyd. of open channels By Bakhmetiff B.A. (McGraw Hill, New York)
6. Open Channel Hyd. By Chow V.T. (McGraw Hill, New York)

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject – Cost Effective & ECO-Friendly Structures

Sub Code – CE 74 (B)

Elective-I

UNIT –I

Concepts of energy efficient & environment friendly materials and techniques. Cost effective materials :- Soil, Fly ash, Ferrocement, Lime, Fibres, Stone Dust, Red mud, Gypsum, Alternate Wood, Polymer.

Energy Efficient & Environment friendly building material products:- Walls - Stabilised and sun dried, soil blocks & bricks, Solid & Hollow concrete blocks, stone masonry blocks, Ferrocement partitions.

Roofs - Precast R.C. Plank & Joists roof, Precast channel roof, Precast L-panel roof, Precast Funicular shells, Ferrocement shells, Filler Slab, Seasal Fibre roof, Improved country tiles, Thatch roof, M.C.R. tile.

UNIT -II

Cost effective construction techniques and equipments :-

1. Techniques :- Rat trap bond construction, Energy Efficient roofings, Ferrocement technique, Mud Technology.
2. Equipments :- Brick moulding machine, Stabilised soil block making machine and plants for the manufacturing of concrete blocks, M.C.R. tile making machine, Ferrocement wall panel & Roofing channel making machine, R.C.C. Chalkhatmaking m/c.

UNIT -III

Cost effective sanitation :-

1. Waste water disposal system
2. Cost effective sanitation for rural and urban areas
3. Ferrocement Drains

UNIT -IV

Low Cost Road Construction:-

Cost effective road materials, stabilization, construction techniques tests, equipment used for construction, drainage, maintenance.

UNIT-V

Cost analysis and comparison :-

1. All experimental materials
2. All experimental techniques- Green Building rating systems

Reference books:-

1. Alternative Building Materials and Technologies – K S Jagadeesh, B V Venkatta Rama Reddy & K S NanjundaRao – New Age International Publishers
2. Integrated Life Cycle Design of Structures – AskoSarja –CRC Press
3. Non-conventional Energy Resources –D S Chauhan and S K Sreevasthava – New Age International Publishers
4. Buildings How to Reduce Cost – Laurie Backer - Cost Ford
5. Lynne Elizabeth, Cassandra Adams Alternative Construction : Contemporary Natural BuildingMethods ”, Softcover, Wiley & Sons Australia, Limited, John,2005
6. Givoni, “Man, Climate, Architecture, Van Nostrand, New York, 1976.
7. Charles J. Kibert, Sustainable Construction: Green Building Design and Delivery, John Wiley & Sons,2005.
8. Eugene Eccli- Low Cost, Energy efficient shelter for owner & builder, Rodale Press, 1976

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject – Advance Pavement Design

Sub Code – CE 74 (C)

Unit -I

Equivalent Single Wheels Load concepts and applications, Relationship between wheel arrangements and loading effects, tyre contact area, Effect of load repetition, Effect of transient loads, Impact of moving loading, Factors to be considered in Design of pavements, Design wheel load, soil, climatic factors, pavement component materials, Environmental factors, Special factors such as frost, Freezing and thawing.

Unit -II

Flexible Pavements : Component parts of the pavement structures and their functions, stresses in flexible pavements, Stress distribution through various layers, Boussinesque's theory , Burmister's two layered theory, methods of design, group index method, CBR method, Burmister's method and North Dakota cone method.

Unit -III

Rigid Pavements: Evaluation of subgrade, Modulus-K by plate bearing test and the test details, Westergaard's stress theory stresses in rigid pavements, Temperature stresses, warping stresses, frictional stresses, critical combination of stresses, critical loading positions.

Unit -IV

Rigid pavement design: IRC method, Fatigue analysis, PCA chart method. AASHTO Method, Reliability analysis. PAVEMENT JOINTS: Types of joints, contraction and warping joints, dowel bars and tie bars, Temperature reinforcements, filling and sealing of joints.

Unit -V

Evaluation and Strengthening of Existing Pavements: Benkleman beam method, Serviceability Index Method. Rigid and flexible overlays and their design procedures.

Reference Books:--

1. Principles of pavement design by E.J.Yoder & M.W. Witczak
2. AASHO, "AASHO Interim Guide for Design of Pavement Structures", Washington, D.C.
3. Portland Cement Association, Guidelines for Design of Rigid Pavements, Washington,
4. DSIR, Conc. Roads Design & Construction
5. Srinivasan M. "Modern Permanent Way"

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject – Intellectual Property Rights

Sub Code – CE 74 (D)

Elective-I

UNIT I - Introduction

Introduction and Justifications of IPR, Nature of IP, Major forms of IP- Copyright, Patent, Trade Marks Designs, Geographic indication, layout design of Semi conductors, Plant varieties, Concept & Meaning of Intellectual Property.

Major international documents relating to the protection of IP - Berne Convention, Paris Convention, TRIPS. The World Intellectual Property Organization (WIPO).

UNIT II - Copyright

Meaning and historical development of copyright , Subject matter , Ownership of copyright, Term of copyright, Rights of owner, Economic Rights, Moral Rights. Assignment and licence of rights, Infringement of copyright, Exceptions of infringement, Remedies, Civil, Criminal, Administrative, Registration Procedure.

UNIT III - Patents

Meaning and historical development,. Criteria for obtaining patents, Non patentable inventions, Procedure for registration, Term of patent, Rights of patentee, Compulsory licence, Revocation, Infringement of patents, Exceptions to infringement, Remedies, Patent office and Appellate Board.

UNIT IV – Trade Marks, Designs & GI

Trade Marks: Functions of marks, Procedure for registration, Rights of holder, Assignment and licensing of marks, Infringement, Trade Marks Registry and Appellate Board.

Designs: Meaning and evolution of design protection, Registration, Term of protection, Rights of holder, unregistered designs.

Geographical Indication: Meaning and evolution of GI, Difference between GI and Trade Marks, Registration, Rights, Authorised user.

UNIT V Contemporary Issues & Enforcement of IPR

IPR & sustainable development, The Impact of Internet on IPR. IPR Issues in biotechnology, E- Commerce and IPR issues, Licensing and enforcing IPR, Case studies in IPR

References:

1. P. Narayanan, Intellectual Property Law, Eastern Law House
2. Neeraj Pandey and Khushdeep[Dharni, Intellectual Property Rights, PHI, 2014
3. N.S Gopalakrishnan and T.G. Agitha, Principles of Intellectual Property, Eastern Book Co. Lucknow, 2009.
4. Anand Padmanabhan, Enforcement of Intellectual Property, Lexis Nexis Butterworths, Nagpur, 2012.
5. Managing Intellectual Property The Strategic Imperative, Vinod V. Sople, PHI.
6. Prabuddha Ganguli, “ Intellectual Property Rights” Mcgraw Hill Education, 2016.

BHABHA UNIVERSITY, BHOPAL

Branch- Civil Engineering

Subject – Modern Construction Technique & Equipment

Sub Code – CE 75

Unit I

Modern Construction Materials: Study of advance building materials like, aluminum, glass, fabric, various types of finishes & treatments, construction chemicals – sealants, engineering grouts, mortars , admixtures and adhesives

Unit II

Polymers in civil engineering-structural plastics and composites- polymer membranescoatings-adhesives, non - weathering materials-flooring and facade materials- glazed brick, photo catalytic cement, acid etched copper and composite fiber metals-metals and special alloys of steel - water jet cut stainless steel, mill slab steel, tension rods assemblies and cast iron, heat treatment in steels, tendons.

Unit III

Construction methods: precast flat panel system, 3d volumetric construction, tunnel boring methods, slip form work, precast foundations .fabrication of pre cast and pre stressed components, reinforcing steel: types, bending, placing, splicing and spacing, tendons- soil improvement - mechanical, thermal and chemical.

Unit IV

Construction Equipment's: equipment for excavating, dredging, trenching, tunneling, drilling, blasting-equipment for compaction-erection equipment- types of pumps used in construction-equipment for dewatering and grouting-foundation and pile driving equipment , forklifts and related equipment-portable material - conveyors-hauling equipment.

Unit V

Smart Materials: Concept and types, sensing technology-types of sensors –physical measurement using piezoelectric strain measurement, piezoelectric and electrostrictive material - magneto structure material, shape memory alloys, electro rheological fluids.

References Books:

1. Shan Somayaji, Civil Engineering Materials 2nd Edition, Prentice Hall Inc., 2001.
2. Mamlouk M.S. and Zaniewski J.P., Materials for Civil and Construction Engineers, Prentice Hall Inc., 1999.
3. Derucher K., Korfiatis G. and Ezeldin S., Materials for Civil and Highway Engineers ", Prentice Hall Inc., 1999. 4th Edition
5. Sharma S.C. Construction Equipment and Management, Khanna Publishers New Delhi, 1988.
6. Deodhar S.V. Construction Equipment and Job Planning, Khanna Publishers, New Delhi, 1988.
9. Mukesh V. Gandhi, Brian S. Thompson, Smart Materials and Structures, Springer,

Practical work:

1. Study of basic properties and tests on modern materials
2. Collect the specification of various modern construction materials and equipment available in market
3. Prepare and give a presentation on any of the topic content in syllabus.